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1. Document

This work illustrates one way to tackle a machine learning task. In particular, this example is intended as a guidepost towards more reliable conclusions from machine learning through better methodology.

The document focuses on methodological reasoning and interpretation, while implementation details can be found in the Python script.

1.1 Introduction

When applying machine learning, the first and most fundamental question is: what exactly are we trying to achieve? Here, we analyse the Wisconsin Breast Cancer Dataset with two objectives: distinguishing as accurately as possible between malignant and benign tumours, and identify, interpretably how to reach this distinction. We also want the models to be deterministic.

1.2 Methodology

Before training any model, it is essential to understand the structure and quality of the dataset. While it is technically possible to apply machine learning algorithms without deeper inspection, doing so is not advisable. Even well-curated datasets require addressing several questions:

- Are any features obviously irrelevant (for example, an ID column)?
- What types of variables are present: numerical, categorical, or mixed?
- Are there missing values?
- Are there extreme or implausible values that might indicate errors?

This stage introduces the greatest degree of freedom in the entire workflow, as decisions do not and cannot rely on objective ground truth. For the present dataset, the situation is relatively straightforward though. There are no clearly irrelevant columns (the scikit-learn version already separated »ID« and »Diagnosis«), all variables are numeric, there are no missing values, and there is no indication of faulty or implausible measurements.

Nevertheless, the topic of outliers deserves explicit discussion. Outliers may arise from several sources. First, they may occur when individuals outside the intended population are included in the dataset. For example, non-English speakers participating in the validation of an English vocabulary test. Importantly, such observations are not necessarily incorrect; they may accurately reflect the participants' characteristics but still be inappropriate for the research question. Second, outliers may represent legitimate observations that provide valuable information about relationships in the data, such as

individuals who genuinely possess an unusually high or low level of the trait being measured. Third, they may arise from extreme random error, such as a participant guessing all answers correctly on a multiple-choice test. Finally, they may result from mistakes in measurement, recording during data collection, or errors introduced during data preparation [1].

In general, observations ought to be retained unless there is a clear reason to believe that they are erroneous or inconsistent with the research question. Empirical work by Gress et al. [2] supports this caution: they found that removing a single pair of outliers had measurable effects on Type I error rates even in datasets with thousands of observations. The effect can be even larger when removed observations lie farther from the central tendency.

1.2.1 Model Selection

Given the dataset's characteristics (numerical features and a binary classification) both objectives can be achieved using Support Vector Machines with a linear kernel and Logistic Regression. These models support interpretability through linear decision boundaries and feature weights, are largely deterministic (when configured correctly), and have demonstrated solid performance across classification tasks as shown by [3]–[5].

1.2.2 Training and evaluation Procedure

The goal is to minimize bias, obtain statistically valid results, and remain computationally practical.

Rather than relying on a single random seed, experiments are repeated across multiple runs with different random initialisations and data splits. Because, evaluating a model on a single partition can yield overly optimistic or pessimistic estimates if that split is unusually favourable or unfavourable. Repeated runs mitigate this risk by ensuring that conclusions do not depend on any particular configuration.

Model performance is estimated using repeated stratified k-fold cross-validation, preventing folds from being accidentally unrepresentative. For each split, models are trained on the training folds and evaluated on the corresponding test fold, with performance metrics recorded for every evaluation.

Each classifier is implemented as a pipeline in which feature standardisation is applied within each training fold before model fitting. This step is particularly important for models such as support vector machines and logistic regression, whose optimisation are sensitive to feature scale. Standardisation parameters are estimated exclusively from the training portion of each fold and subsequently applied

to the corresponding test data. This ensures that no information from the evaluation data influences model training, preventing data leakage.

Each pipeline corresponds to a predefined model configuration with fixed hyperparameters, and no hyperparameter optimisation is performed within the cross-validation procedure itself. This design avoids the form of selection bias described by Cawley and Talbot [6], which arises when model configurations are tuned and evaluated using the same resampling procedure. Consequently, the resulting performance distributions provide approximately unbiased estimates for each fixed configuration.

1.2.3 Statistical Framework

This raises the question of how conclusions are drawn from experimental results. Null hypothesis significance testing (NHST) has been customarily adopted in machine learning, but its limitations are widely discussed across scientific disciplines. A key issue is the discrepancy between what researchers expect from a significance test and what it actually delivers. Consider this example: we test whether a treatment changes task performance and obtain a significant result ($p = 0.01$). Evaluate the following statements [7], [8]:

- You have determined the probability that the null hypothesis is true.
- You can infer the probability that the alternative hypothesis is true.
- The result is reliable in the sense that 99 % of replications would also be significant.

The p-value is defined as the probability of observing the data (or more extreme data) under the assumption that the null hypothesis is true. So which statements are actually true? In fact, all three statements are false. A significance test does not provide the probability that a hypothesis is true, and statement three would only be valid if the null hypothesis were known to be true. Statements such as »with 90 % confidence method A outperforms alternatives« are similarly misleading. A 90% confidence interval refers to long-run coverage properties under repeated sampling, not to the probability that method A is better. Yet these are precisely the questions we care about. They can be addressed using Bayesian methods however [7], [8].

1.2.4 Bayesian Bootstrap

The method used here is the Bayesian bootstrap, introduced by Rubin [9]. In practice, this corresponds to placing a uniform Dirichlet prior over the weights assigned to the observed samples, and repeatedly drawing such weightings to approximate the posterior distribution.

To estimate the performance of each model individually, samples are drawn independently for each model. This produces stable estimates of the expected mean performance. From these samples we compute a point estimate (the posterior mean) and a credible interval. Unlike a classical confidence interval, a credible interval has a direct probabilistic interpretation: given the data, there is an X % probability that the true mean lies within the interval.

A related question is which model is most likely to be the best. When all models are evaluated on the same data splits, their scores are correlated. To account for this, the bootstrap draws a single set of weights that is applied to all models simultaneously. This preserves the relationships between their scores. For each draw we identify the model with the highest weighted mean, and by repeating this many times we estimate the probability that each model is the best-performing one.

A related limitation concerns the identification of a top-performing model. When selecting the best model from a finite set of estimates, the winner's expected performance is slightly inflated - not due to overfitting in the model selection procedure, but because selecting the maximum from noisy estimates introduces an upward bias (Winner's Curse). This directly affects the bootstrap probability estimates above: Both the ranked probabilities and the reported performance of the top-ranked model should be interpreted as conditional on this specific configuration set and resampling procedure, with the true expected performance likely lying slightly below the posterior mean estimate.

1.2.5 Region of Practical Equivalence

This probability is refined by considering a ROPE (Region of Practical Equivalence). Rather than asking which model has the strictly highest posterior mean - a question sensitive to arbitrarily small differences we ask which model is practically best: ahead of all others by more than some threshold. A model within the ROPE of the best is treated as effectively equivalent. Importantly, the ROPE is not a statistical parameter but a utility decision. Setting ROPE = ± 0.01 asserts that performance differences smaller than one percentage point are practically negligible.

A threshold around 1 percentage point is a reasonable starting point: empirical analysis by Wainer [3] suggest this is roughly the median change observed when evaluating algorithms on new datasets drawn from the same population.

Smaller ROPE values may be appropriate when marginal gains carry meaningful clinical or operational benefits; larger values may be justified when deployment variability or dataset shift dominates small improvements. However, rather than committing to a single value, we evaluate multiple ROPE values as a form of sensitivity analysis.

1.2.6 Evaluation Metrics

We now have a defined dataset & objective, selected models, a training procedure, and a statistical framework. Finally, we must define which performance metrics to use.

In medical diagnosis, like in this case, identifying true positive cases is the highest priority. The relevant quantity is the true positive rate:

$$TPR = \frac{TP}{TP + FN}$$

However, this metric alone is insufficient. A classifier that always predicts »positive« achieves a TPR of 100 % while performing no meaningful discrimination. We must therefore also consider how many true negatives are incorrectly labelled positive / how many true negatives are correctly labelled negative - the true negative rate:

$$TNR = \frac{TN}{TN + FP}$$

Together, TPR and TNR describe how well a model separates the two classes, but they do not indicate how reliable a positive prediction is once made. This becomes critical with class imbalance. Consider 1,000 patients of whom 10 are truly diseased. Suppose a model correctly identifies all 10 (TP = 10, FN = 0) but produces 190 false positives (TN = 800). Then TPR = 100 % and TNR = 80 %. Yet of the 200 patients flagged positive, only 10 are actually diseased. The positive predictive value being:

$$PPV = \frac{TP}{TP + FP} = 5\%$$

The negative predictive value being:

$$NPV = \frac{TN}{TN + FN} = 100\%$$

These four metrics can actually be condensed into two combined measures:

$$\begin{aligned} \text{Informedness (Youden's J):} &= TPR + TNR - 1 \\ \text{Markedness (deltaP):} &= PPV + NPV - 1 \end{aligned}$$

As such, Informedness captures how well the model separates true classes (ground-truth perspective), while markedness captures how trustworthy its predictions are (prediction perspective). In the 1,000-patient example: Informedness = 0.8, Markedness = 0.05.

There is a large body of literature assessing metrics (like [10]–[12]) and the only thing certain is that there isn't a silver bullet, but there are reasons to believe TPR, and Informedness are good metrics to use, here and in general, with Foulle [13] having described an Informedness extension for more than 2 classes.

Not all datasets require equal emphasis on both however. For the present dataset, markedness is comparatively less critical. The maximum possible divergence between the two measures, given that informedness exceeds markedness, is approximately 0.0526 (see Appendix). This constraint is not particularly limiting here, since our objective is to maximize correct classifications; a low informedness value would render a model unsuitable regardless of markedness. Datasets with stronger class imbalance require closer attention to prediction reliability, but in this case the difference between informedness and markedness is limited. As such we can focus on Informedness alone, which simplifies things not only on a methodological level but it also reduces mental overhead when interpreting the results.

1.3 Results

Both models achieve very similar predictive performance under repeated stratified cross-validation. Across repeated partitions, both models can achieve informedness values around 0.94-0.95, and Bayesian comparison indicates no meaningful difference between them across the ROPE values considered. These cross-validated results provide an estimate of the classifiers' expected performance on unseen data drawn from the same population.

To examine the structure of the learned decision rules themselves, selected configurations were subsequently refitted on the full dataset and their decision scores analysed. Both models identify medically meaningful patterns: features associated with tumour size, irregular texture, and concavity receive the largest absolute coefficients. The similarity in feature rankings across models suggests that these relationships reflect genuine structure in the data rather than artefacts of a particular algorithm.

However, false negatives warrant particular attention: malignant tumours misclassified as benign represent the most clinically consequential error type, as they risk delaying diagnosis and treatment. The confidence of these misclassifications suggests atypical presentations outside the training distribution, rather than simply borderline cases. Further investigation into the characteristics of these samples is strongly recommended, as they could reveal systematic gaps in the training data or highlight patient subgroups that are underrepresented in the present dataset.

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Appendix

Maximum Divergence Between Informedness and Markedness

The Wisconsin dataset has 212 malignant and 357 benign cases, giving:

$$TP + FN = 212, \qquad TN + FP = 357.$$

We can therefore express FN and FP purely in terms of TP and TN:

$$FN = 212 - TP, \qquad FP = 357 - TN.$$

The four metrics of interest are:

$$TPR = \frac{TP}{212}, \quad TNR = \frac{TN}{357}, \quad PPV = \frac{TP}{TP + (357 - TN)}, \quad NPV = \frac{TN}{TN + (212 - TP)}.$$

Informedness and Markedness are then:

$$J = TPR + TNR - 1 = \frac{TP}{212} + \frac{TN}{357} - 1,$$

$$\Delta P = PPV + NPV - 1 = \frac{TP}{TP + 357 - TN} + \frac{TN}{TN + 212 - TP} - 1.$$

Substituting $x = TP$ and $y = TN$, we define the objective:

$$f(x,\, y) = J - \Delta P = \frac{x}{212} + \frac{y}{357} - \frac{x}{x + 357 - y} - \frac{y}{y + 212 - x},$$

which we maximise subject to:

$$x \in [0, 212], \quad y \in [0, 357],$$

$$f(x, y) > 0 \quad (\text{i.e. } J > \Delta P),$$

$$J(x, y) > 0.$$

Finding the maximum. Solving $\nabla f(x, y) = 0$ yields no solution in the interior of the feasible region. By the Karush–Kuhn–Tucker conditions, any optimum must therefore occur on the boundary of the domain or where constraints are active.

Under these constraints, the maximum occurs at:

$$(x, y) = (212, 294).$$

yielding:

$$J = \frac{212}{212} + \frac{294}{357} - 1 = \frac{294}{357} \approx 0.8235,$$

$$\Delta P = \frac{212}{212 + 357 - 294} + \frac{294}{294 + 212 - 212} - 1 = \frac{212}{275} + \frac{294}{294} - 1 = \frac{212}{275} \approx 0.7709,$$

$$\boxed{f_{\max} = J - \Delta P \approx 0.8235 - 0.7709 = 0.0526.}$$

But we could also just skip all this mathematical proving and go with the brute-force method:

```
max_f = float("-inf")
best_cases = []

for TP in range(213):
  FN = 212 - TP

  for TN in range(358):
    FP = 357 - TN

    if TP + FP == 0 or TN + FN == 0:
      continue

    TPR = TP / 212
    TNR = TN / 357
    PPV = TP / (TP + FP)
    NPV = TN / (TN + FN)
    J = TPR + TNR - 1
    deltaP = PPV + NPV - 1

    f = J - deltaP

    if f <= 0:
      continue
    if J <= 0:
      continue

    if f > max_f:
      max_f = f
      best_cases = [(TP, TN, FP, FN, f)]
    elif abs(f - max_f) < 1e-12:
      best_cases.append((TP, TN, FP, FN, f))

print("Max f:", max_f)
for case in best_cases:
  print(case)
```